

Pay, Oscillate, or Learn?

Three Agent Decision Rules under Congestion Pricing

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Agent-Based Modelling

Q-Learning

El Farol

Auckland CBD

ToU Pricing

Why Does the Behavioural Assumption Matter?



Auckland's Context

Congestion costs NZ\$2.6B/yr. Time-of-Use (ToU) charging legislation passed Nov 2025. CBD cordon launch expected alongside City Rail Link.



The Gap

Most models ask 'how much will traffic fall?' — but behavioural assumptions drive whether congestion shifts in time, in space, or not at all.



Our Approach

ABM with 2,500 heterogeneous agents (1 agent = 160 vehicles, calibrated to observed daily counts), Auckland CBD road network. Three decision rules, same pricing environment, same LoS measurement.

Three Agent Decision Rules: A Rationality Spectrum

← Static Rationality

01

Exponential Decay

Static Rationality

$$P(\text{entry}) = e^{(-\alpha \cdot \text{fee} / \text{VoT})}$$

- Entry probability declines with fee/VoT ratio
- No memory, no adaptation
- Standard equilibrium assumption
- Result: peak V/C -12% (fixed hour) to -26% (may reroute)

02

El Farol Bar Model

Bounded Rationality (No Learning)

Enter if best predictor: $E[\text{congestion}] < \theta$

- Portfolio of predictors based on recent history
- Reads yesterday's congestion, not today's fee
- Near-universal entry (91%), so little left to deter
- Result: 0% in every arm — the charge never binds

Bounded Rationality with Learning →

03

Q-Learning

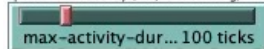
Bounded Rationality with Learning

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$$

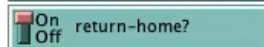
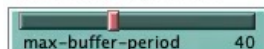
- State = time band x congestion band; action = enter or not
- Fee is absent from the decision — it enters only via the reward
- So the response is learned over days, not applied at once
- Result: -39% if staying home is the only option, -14% if not



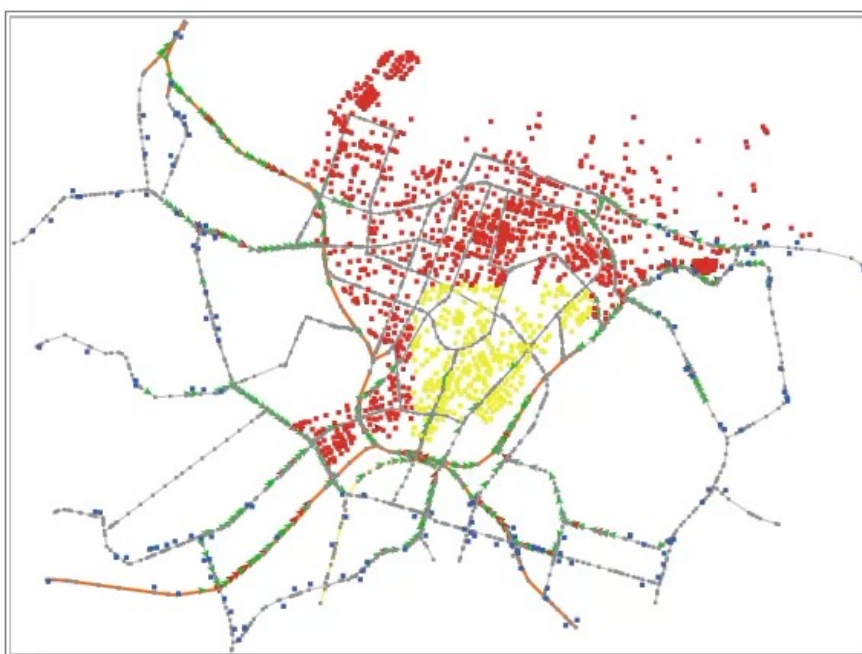
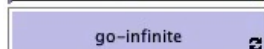
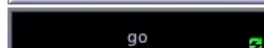
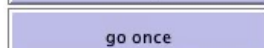
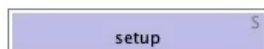
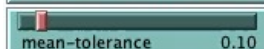
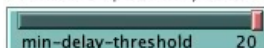
Maximum number of ticks a vehicle spends at an activity (i.e., at a building)



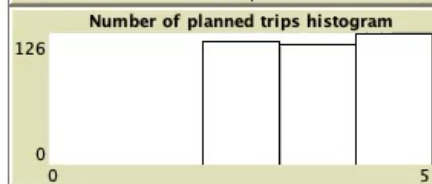
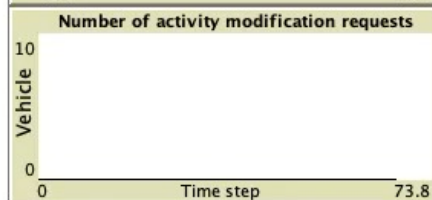
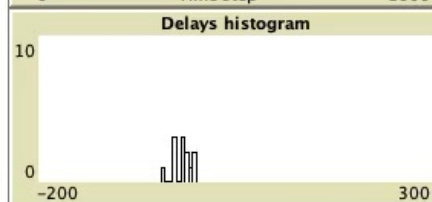
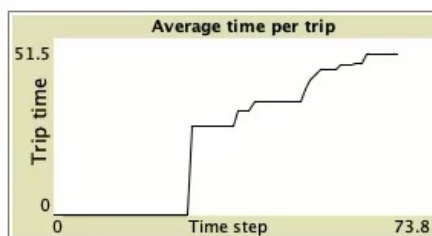
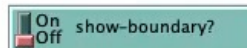
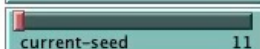
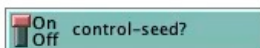
Maximum number of ticks that the vehicle moves earlier than it should to reach its next building destination



Minimum delay threshold at which an activity modification is requested is only reached if w



Control the seed value (Note that go-infinite increments the current-seed)



What Happens When You Charge Drivers?

Same road. Same price. Three very different outcomes — depending on how drivers think.



Baseline Driver

"I check the price and decide."

Fewer drivers enter at the peak: entry falls from 52% to 40%. Give them an hour of flexibility and the cut deepens from 12% to 18%.

Peak V/C (inner cordon)

 No charge 0.113

 ToU 0.100

↓ Steady reduction



Herd Driver

"Everyone avoided it yesterday, so I'll go today."

Drivers read yesterday's congestion, not today's price. Almost everyone still enters (91%), so the charge has nothing left to deter — in every arm we tried.

Peak V/C (inner cordon)

 No charge 0.170

 ToU 0.170

→ No response



Learning Driver

"The fee stung last time — I'll skip the trip today."

Drivers learn day by day. But this -39% assumes the only alternative is staying home: let them shift an hour and it falls to -14%.

Peak V/C (inner cordon)

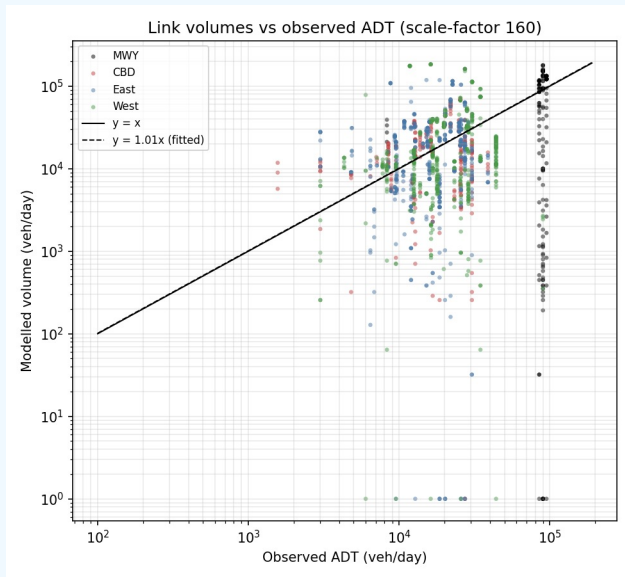
 No charge 0.129

 ToU 0.079

↓ Largest cut in entries

Key insight: the right behavioural assumption changes not just how much congestion falls — but whether the policy works at all.

Calibrated on Volume and Space — Not on Time or OD



Volume: 1 agent = 160 vehicles

Flow-weighted modelled/observed ratio 1.013 across all 1,634 links (median 1.010).
~400,000 vehicle-trips a day.

Space: 1,400 suburban trip ends added

Every non-home destination used to sit inside the cordon. Group ratios went from 0.75–1.80 to 0.82–1.20; CBD from 1.80 to 1.12.

Time: assumed, never fitted

Departure hours come from a fixed weight profile (~20% of trips at 08:00), not from observed data. Implied design-hour factor 0.157 vs the 0.10 assumed, so absolute LoS reads pessimistically.

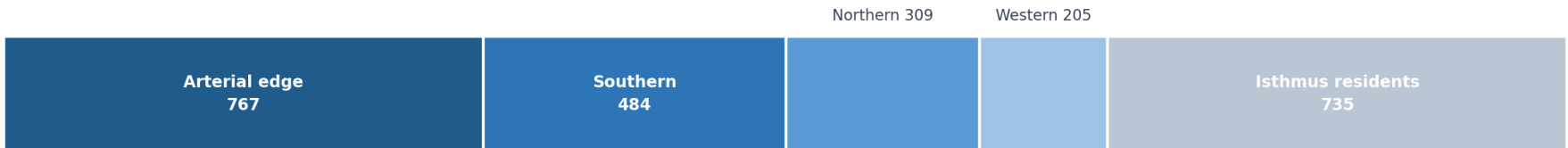
OD matrix: synthetic, not estimated

Origins follow 2023 Census sector shares; destinations are drawn uniformly from the building stock. No gravity model, no observed trip ends. $k = 0.08/0.10/0.12$ leaves direction and ordering unchanged.

So: the model reproduces observed traffic volume and its spatial spread, not observed travel patterns. Calibration still changed the answers — it removed El Farol's oscillation and cut every congestion level, superseding the pre-calibration numbers.

Who Are the 2,500 Agents?

2,500 agents · 1 agent = 160 real vehicles · ~400,000 vehicle-trips a day



Where they live

0 agents live inside the cordon



Whether the charge can reach them

Measured at setup, seed 11. Every agent lives outside the cordon, so the charge is always a decision to cross a boundary — never a resident's parking decision.

The 1,000 agents that never enter are an unpriced comparison group inside the same run.

What Each Agent Carries — Three Real Examples

Three real agents, one charge: $p = \text{base-trip-rate} \times \exp(-\beta \times \text{fee} \div \text{VoT})$, with β itself set inversely to VoT



LOWEST VALUE OF TIME

NZ\$1.34 /h

price sensitivity $\beta = 2.00$ (capped)

essential prob 0.15 · departs 21:00

\$6 fee = 4.48 hours of own time



MEDIAN VALUE OF TIME

NZ\$9.67 /h

price sensitivity $\beta = 0.50$

essential prob 0.08 · departs 09:00

\$6 fee = 0.62 hours of own time



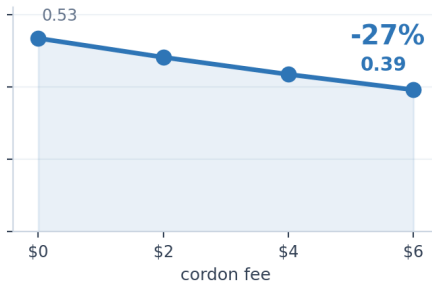
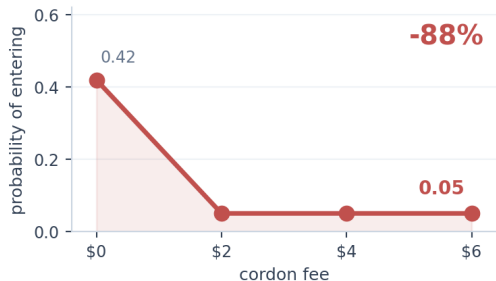
HIGHEST VALUE OF TIME

NZ\$68.79 /h

price sensitivity $\beta = 0.10$ (capped)

essential prob 0.05 · departs 17:00

\$6 fee = 0.09 hours of own time



Every agent carries a value of time, a price sensitivity set inversely to it, a baseline trip rate and an essential-trip probability. Because β scales as $1/\text{VoT}$, the deterrent behaves like $\text{fee} \div \text{VoT}^2$ — halving income roughly quadruples it. That is the whole equity result, visible in three individuals.

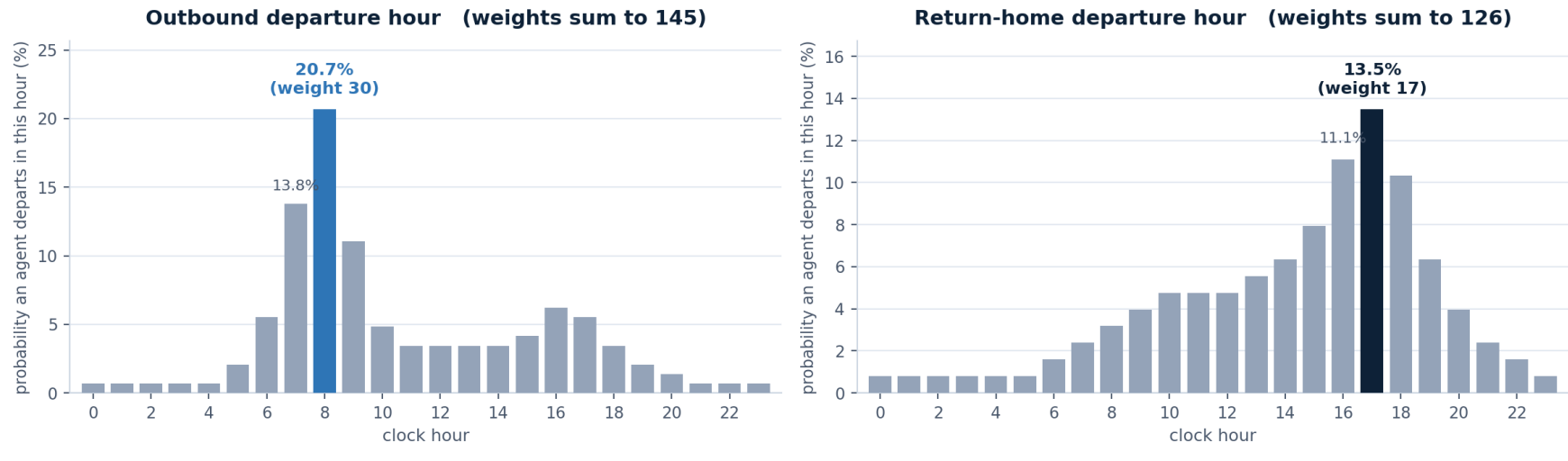
Every Attribute an Agent Carries

ECONOMIC decides the fee response	LEARNING STATE one family per rule	ITINERARY rebuilt every simulated day	MOVEMENT inherited from the base model
vot value of time, NZ\$/h lognormal $\exp(N(2.3, 0.6))$; median 9.67	el-weights, el-scores EI Farol predictor set 5 predictors, weights $U(0, 1)$	v-home home building Census sector shares; none inside the cordon	location, destination position on the network node-to-node along the current link
beta price sensitivity $\text{clip}(0.5 \times \text{medianVoT} / \text{vot}, 0.1, 2.0)$	q-table Q-learning action values 24 states x 4 actions, init $U(-0.1, 0.1)$	b-destinations the day's stops uniform draw, 1-4 plus home; 2.98 mean	speed, top-speed current and free-flow speed BPR-damped by link V/C
base-trip-rate propensity with no charge Uniform(0.3, 0.7); mean 0.50	q-epsilon exploration rate starts 0.40, decays $\times 0.999$ per day	depart-tick, return-tick clock-anchored departure drawn from the hourly weight lists	trip the route being followed cached list of links per OD pair
essential-trip-prob fee-insensitive trip 0.15 / 0.08 / 0.05 by VoT band	q-action action chosen today 0 stay out / 1 travel / 2 earlier / 3 later	trip-hour the hour the fee is charged read off the departure actually made	active?, at-activity? on the road, or stopped active? false ends the agent's day
through? pass-through, never charged 30 % of boundary agents; 544 of 2,500		hour-shift today's retiming -1 / 0 / +1 hour, when retiming is on	tolerance, buffer-period delay tolerance, early start truncated normal / $U(0, \text{max})$

Only the first column decides the response to a charge. The learning state is per rule — an agent uses one family and ignores the others. Distributions are measured at setup, seed 11.

How Departure Hours Are Drawn — Not an OD Matrix

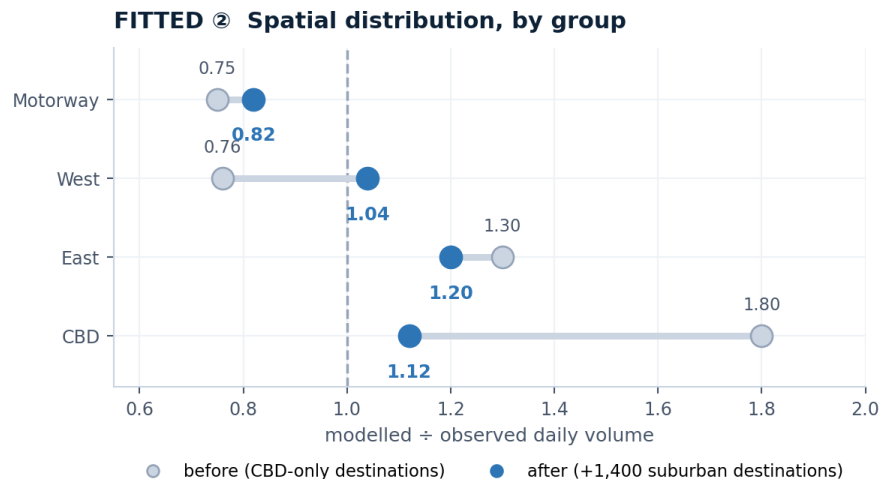
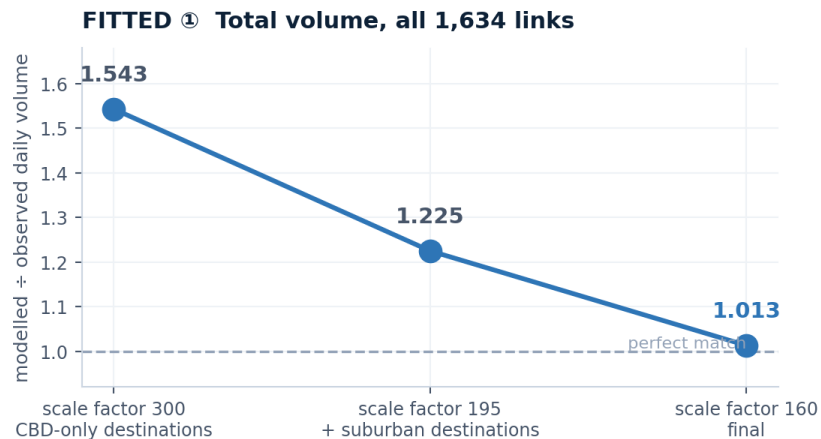
Every agent draws its own hour from these weights, independently, each simulated day — the minute is then uniform within that hour.



Two axes, drawn independently. The OD matrix answers **where** a trip goes; these weights answer **when** it leaves. The weights do not stand in for an OD matrix — they act after the origin and destination are already fixed.

Hourly volume is never assigned. Each of the 2,500 agents spins its own roulette every simulated day, so "how many depart at 08:00" emerges from 2,500 independent draws and differs from day to day. The 20.7 % spike at 08:00 is the direct cause of the too-sharp morning peak (implied $k = 0.157$ against the 0.10 assumed).

What Was Fitted, and What Was Assumed



NOT FITTED: the origin-destination matrix (destinations drawn uniformly) · the time-of-day profile (fixed weight lists)

Two knobs, and only two. The agent scale factor fixes total volume; suburban destinations fix the spatial spread. Both converge on the observed counts.

Nothing fits the temporal axis, which is why the implied design-hour factor sits at 0.157 against the 0.10 assumed — a missing fit, not a poor one.

How Much to Trust Each Number

Evidence status: what is measured, what is only derived, and what is superseded

MEASURED in the runs

Peak V/C, entry trajectories, hourly profiles, LoS shares, the map. Seed 11, 14 days.

DERIVED — not yet measured

The equity result, the OD matrix and the time-of-day profile.

SUPERSEDED — not quoted

All pre-calibration numbers, El Farol's 'oscillation', Learn's -39 %.

RANGES, not point estimates

Pay 12–26 %, Learn 14–39 %, Oscillate 0 % — width set by the action space.

Controlled comparison

One network, one population, one seed across every rule and arm.

Direction robust to capacity assumption

$k = 0.08 / 0.10 / 0.12$ never changes the sign or ordering.

KNOWN BIAS, under repair: a declined CBD entry cancelled the agent's whole day

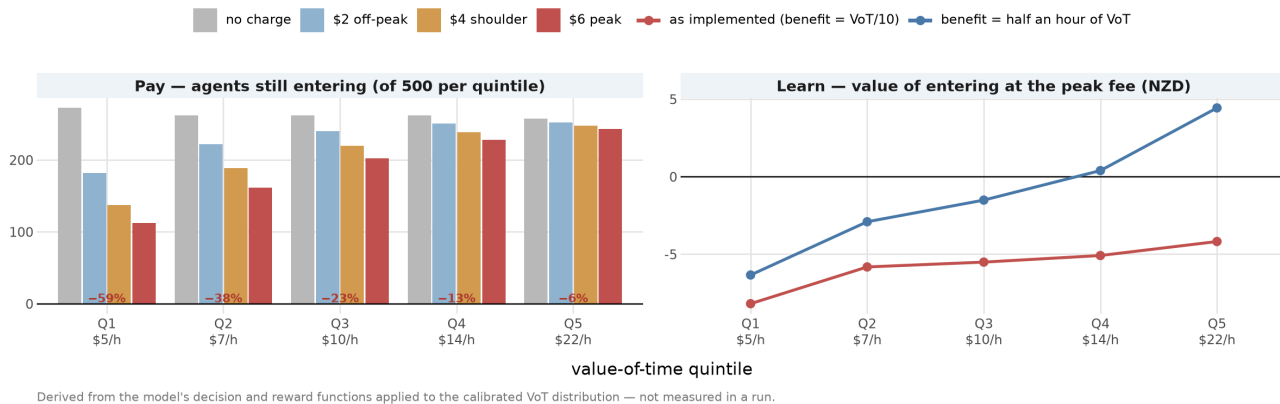
~100,000 suburban vehicle-trips a day vanished with the declining agents, flattering every result outside the cordon. Fixed 29 Jul 2026; these numbers predate the fix.

Who Would Pay? Derived, Not Yet Measured

DERIVED — NOT MEASURED

Who the charge removes from the road

Value-of-time quintile, Q1 = lowest income. The price rule deters the poorest ten times as strongly. The learner deters every band alike, but only because its reward values a commute at a tenth of an hour.



Left (price rule): projected entries per value-of-time quintile as the fee rises. IF the decision rule holds, the \$6 peak fee removes 59% of Q1's trips against 6% of Q5's — the same charge costs Q1 1.30 hours of its own time and Q5 0.28 hours.

Implication, conditional on the rule: most of the congestion relief would be bought by low-income drivers giving up trips.

Right (learner): a diagnosis, not a result — its flat income gradient exists only because the reward prices a commute at VoT/10. Valued at half an hour of VoT, the regressive pattern returns. The learner's equity story is a scaling choice.

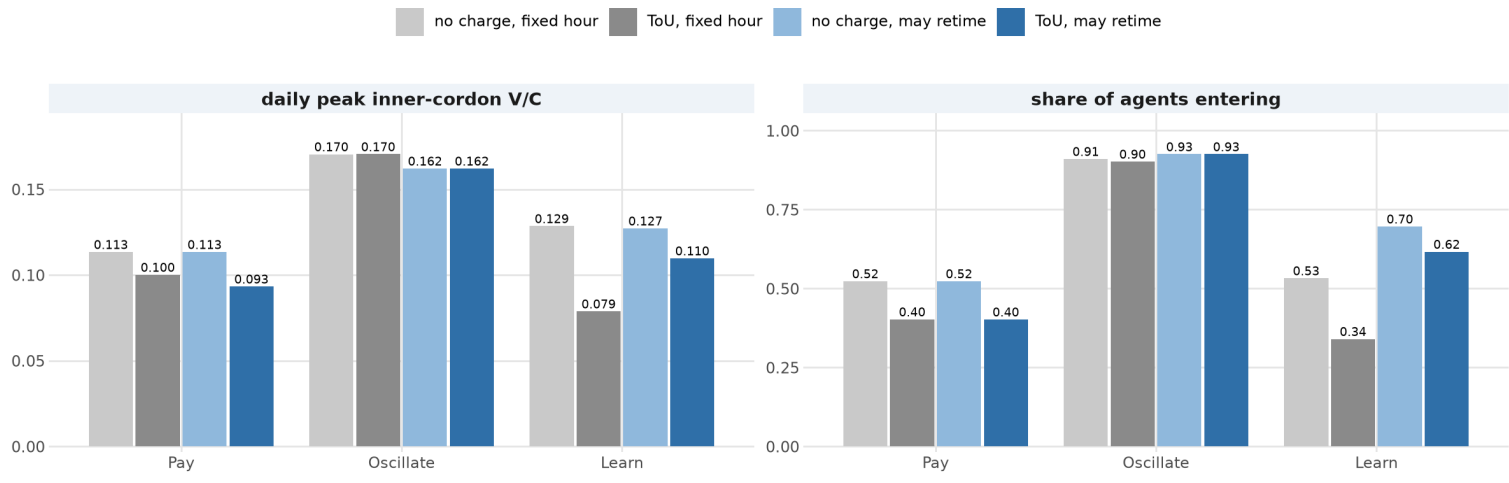
How these numbers were obtained: the model's decision and reward equations applied to the calibrated VoT distribution (median NZ\$10/h). No run has recorded entries by income band, so regressivity is a projection of the assumed rule, not a confirmed simulation outcome. A recording re-run is planned.

Let Them Shift an Hour: Fewer Trips Forgone

Departure-time choice changes how many trips are made

Pay's entry rate is unchanged because only 2.7 % of its entrants shift.

Learn's jumps from 0.34 to 0.62: given somewhere to move, it stops forgoing the trip, and the measured congestion benefit falls with it.



Retiming is the only option that changes HOW MANY trips are made. Pay's entry rate is unchanged (0.40) because only 2.7% of its entrants shift — all of them earlier, all of them low-income, since an hour of schedule delay costs $0.6 \times \text{VoT}$ against a \$2 saving.

Learn is the opposite: given somewhere to move, 67% shift and entry jumps $0.34 \rightarrow 0.62$, so the measured congestion benefit falls from 39% to 14%.

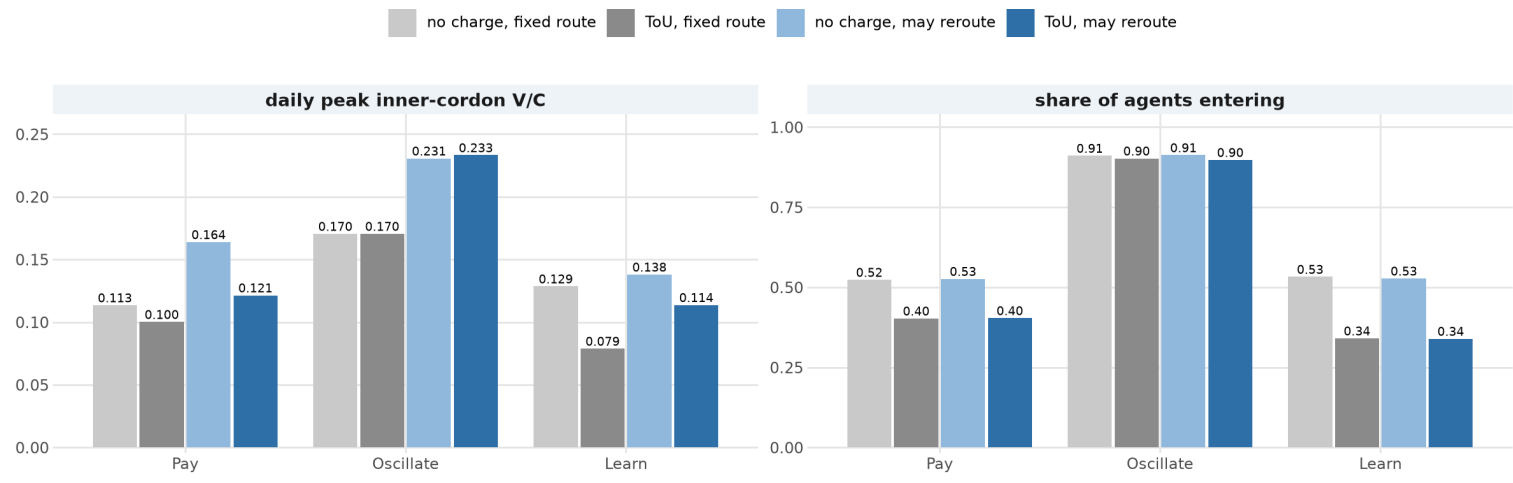
A model where the only response is 'stay home' overstates what the charge achieves.

Let Them Take Another Road: Same Trips, New Places

Route choice changes where the trips go, not how many there are

Entry rates are identical to three decimals in both arms.

What moves is the no-charge baseline: inner V/C rises while the cordon boundary sheds about a third of its load.



Route choice leaves the number of trips untouched — entry rates are identical to three decimals (Pay 0.40, Learn 0.34) — and changes only where those trips go.

The no-charge baseline moves a lot: inner V/C rises (0.113 → 0.164 for Pay) while the cordon boundary falls by a third, because fixed shortest paths were funnelling traffic onto a few approach arterials.

This is the arm that makes displacement testable: the traffic still exists and can move. It still does not pile up outside the cordon.

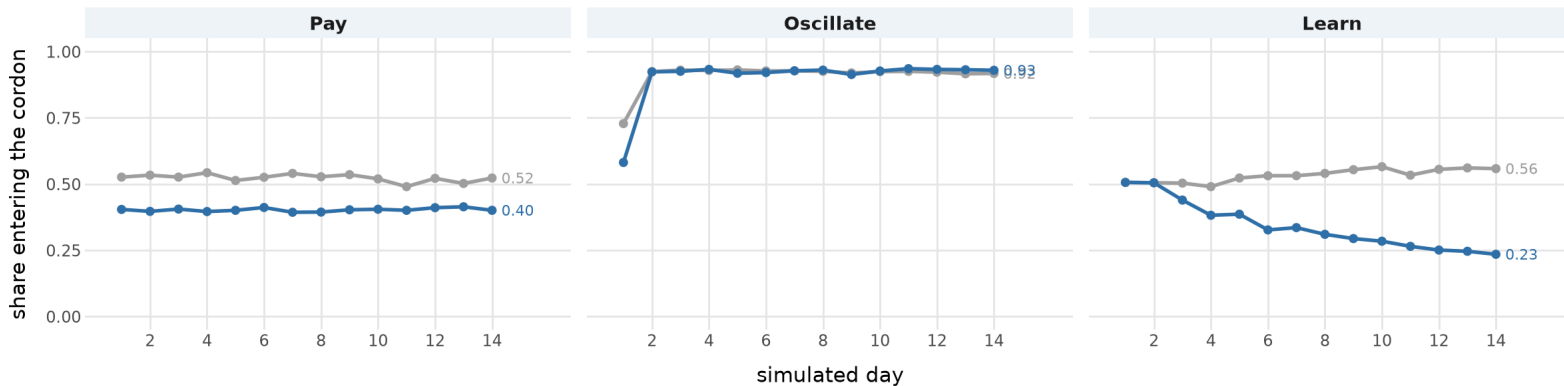
Why Learning Keeps Falling — and Herding Bounces Back

The charge acts immediately under Pay, cumulatively under Learn, and is undone under Oscillate

Share of agents entering the cordon, by simulated day.

Learn starts at the no-charge level because the fee never enters its decision, only its reward.

— No charge — ToU



Pay reads the fee inside the decision, so its response is complete on day 1 and then flat.

Learn never sees the fee when it decides — it only feels it in the reward. Entry starts at the no-charge level and slides for the whole fortnight (0.51 → 0.24), because the fee (\$2–6) outweighs the congestion it could avoid (~0.3 in reward units) by an order of magnitude, so nothing pulls entry back up.

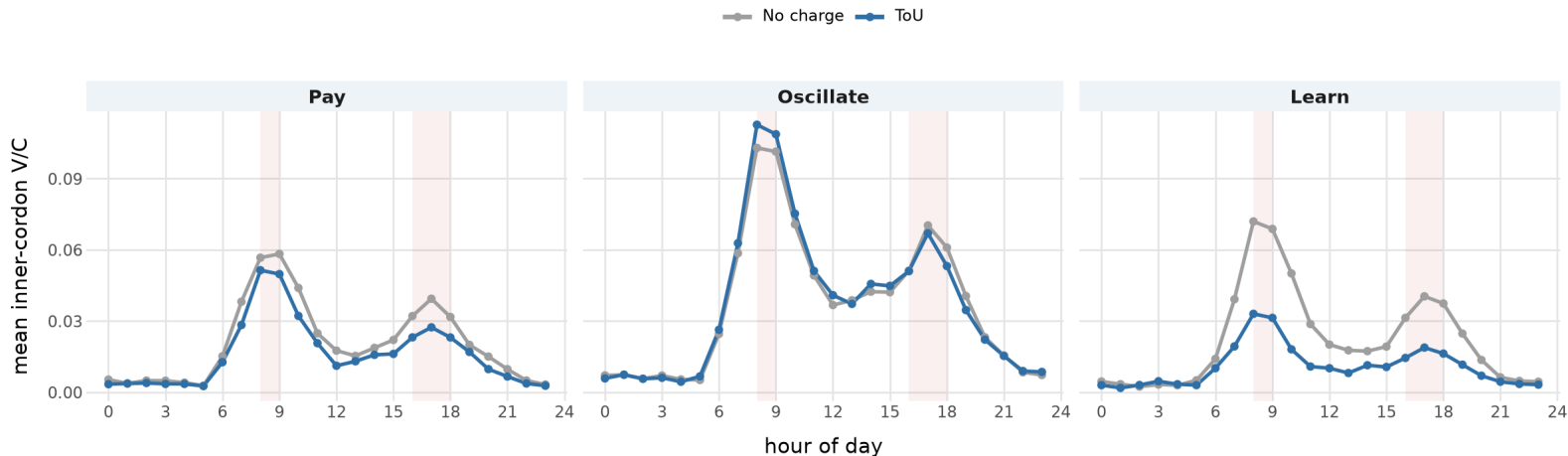
Fee per entering agent is unchanged (2.83 → 2.75): deterrence, not retiming.

Oscillate is deterred on day 1 (0.57 vs 0.73) but its own feedback erases that within a day — by day 2 both regimes sit at 0.91.

Where the Charge Acts Within the Day

Where within the day the charge acts

Mean inner-cordon V/C by hour, days 8-14. Shaded bands are the peak fee windows.
The whole-day mean falls as much as the peaks do, so the charge deters trips rather than spreading them.



Hour-of-day inner-cordon V/C, days 8–14, no charge (grey) vs ToU (blue); shaded bands are the \$6 peak and \$4 shoulder fee windows. Pay –23% at the morning peak, Learn –49%, Oscillate +2%. The whole-day mean falls by as much as the peaks do, so the charge deters trips rather than spreading them.

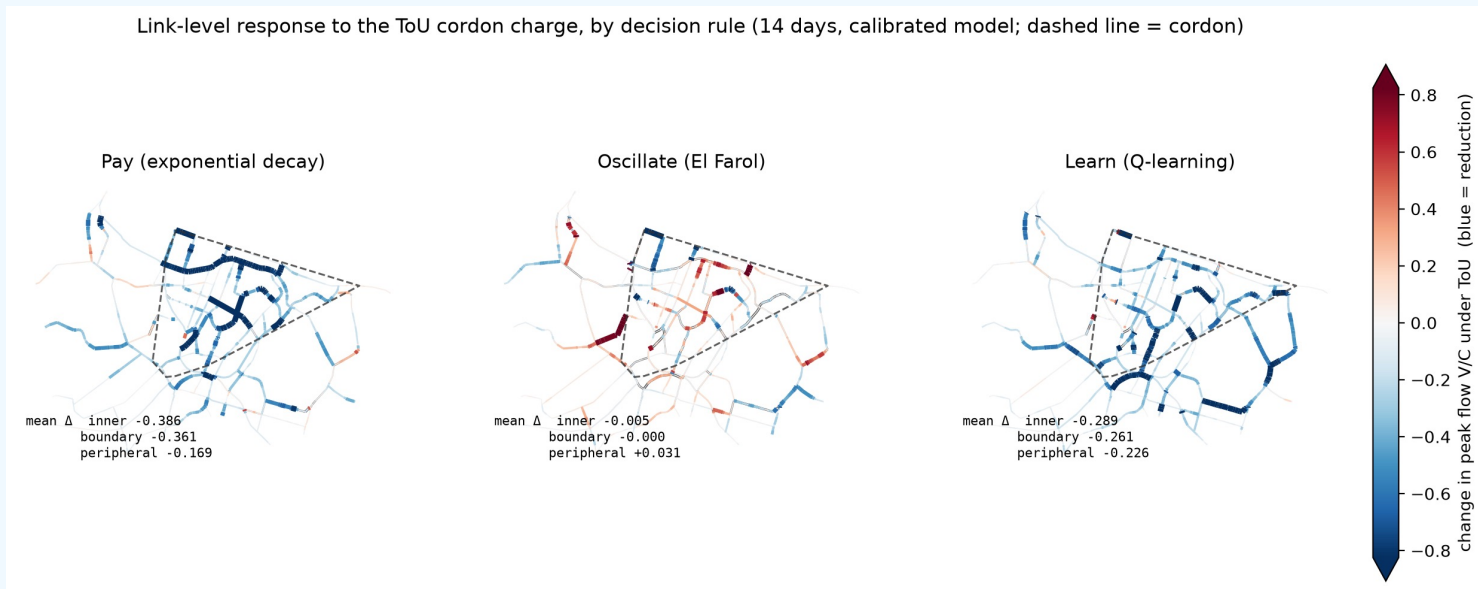
Midday Is Where the Charge Actually Bites

% of each group's traffic at LoS E or worse, no charge → ToU. Mean of days 8–14.

Road group / time band	Pay	Oscillate	Learn
MWY · AM 07–09	53 → 54 (+3%)	62 → 58 (-6%)	53 → 49 (-7%)
MWY · Midday 12–15	46 → 39 (-16%)	62 → 58 (-6%)	48 → 18 (-63%)
MWY · PM 16–18	63 → 60 (-4%)	81 → 82 (+1%)	68 → 58 (-14%)
CBD · AM 07–09	61 → 59 (-4%)	75 → 74 (-2%)	62 → 54 (-14%)
CBD · Midday 12–15	65 → 60 (-7%)	80 → 79 (-1%)	69 → 49 (-30%)
CBD · PM 16–18	70 → 68 (-3%)	83 → 83 (+1%)	72 → 62 (-14%)
East · AM 07–09	69 → 67 (-3%)	77 → 77 (+0%)	70 → 67 (-4%)
East · Midday 12–15	64 → 62 (-3%)	79 → 77 (-3%)	68 → 54 (-20%)
East · PM 16–18	75 → 73 (-2%)	84 → 84 (+0%)	77 → 70 (-9%)
West · AM 07–09	68 → 67 (-0%)	70 → 69 (-1%)	68 → 66 (-3%)
West · Midday 12–15	68 → 64 (-5%)	71 → 68 (-4%)	68 → 61 (-10%)
West · PM 16–18	74 → 75 (+0%)	79 → 79 (-0%)	77 → 76 (-2%)

Green = a fall of 10 % or more, red = the charge made it worse. The peaks barely move because what is left in them is essential travel; the midday shoulder is where discretionary trips sit, and Learn cuts motorway E/F traffic there by 63 %. Oscillate moves nothing anywhere.

No Displacement onto the Ring Roads — So Far



Change in per-link peak V/C under ToU: blue is a reduction, red an increase, dashed outline is the cordon. Under Pay and Learn the boundary and approach roads fall together with the interior (-24% and -37% at the boundary). Under Oscillate the map is mottled and the net change is zero.

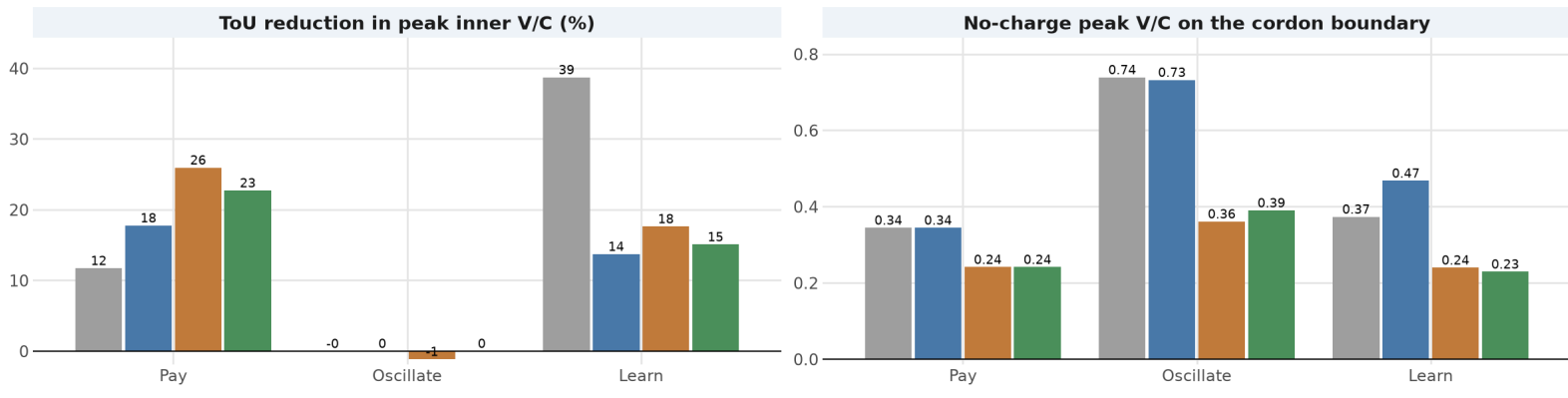
Caveat: in these runs a declining agent's suburban trips were cancelled along with its CBD trip (~100,000 vehicle-trips/day), which flatters the reductions outside the cordon. The model is now fixed; this map awaits a re-run.

What We Let Agents Do Changes the Answer

What we let agents do changes the answer

Same network, population and fee schedule; only the set of permitted responses differs.
Pay is robust at 12-26 %; Learn swings from 39 % to 14 %; Oscillate is zero throughout.

fixed hour, fixed route may retime may reroute both



Same network, same population, same fee schedule. The only difference is the set of responses an agent is allowed: forgo the trip only (grey), or also shift an hour (blue), reroute (orange), or both (green).

Pay is robust at 12–26%. Learn swings 39% → 14% — its headline needs the assumption that staying home is the only alternative. Oscillate is zero in every arm.

Right: routing alone halves the load on the cordon boundary before any charge — the biggest single effect here is a modelling assumption, not a policy.

What This Model Cannot Say

Seven limits that bound every number in this talk

Single seed

Spreads are within-seed. Only El Farol has a five-seed replication.

Fee level untested

No fee-level sweep. The beta sweep varies sensitivity, not price.

Equity is derived, not measured

From the equations, not the runs. No entries recorded by income band.

Learn rests on two arbitrary settings

Reward scale and action space between them decide the headline.

Routing answers to congestion, not the fee

A CBD-bound agent pays whichever road it takes.

Demand is synthetic, not observed

No observed OD matrix or time-of-day profile behind the demand.

A declined trip cancelled the agent's whole day — fixed 29 Jul 2026, re-run pending

Declining deactivated the whole multi-stop agent, so ~100,000 suburban vehicle-trips a day vanished. Fixed 29 Jul 2026; every number here predates the fix.

Conclusions & Next Steps

Key Findings

- 1 Behavioural assumption shapes both magnitude and temporal redistribution of congestion — not just the quantity
- 2 Q-learning's headline depends on what agents may do: -39% when the only option is to stay home, -14% when they can shift an hour
- 3 El Farol shows what happens when agents track each other rather than the fee — entry stays at 91% and the charge does nothing
- 4 ABM places competing theories side-by-side within a common environment — revealing structural differences invisible to aggregate models

Next Steps

Model Extension

Mode choice, household VoT calibration from NZ travel survey data

LLM-ABM Hybrid

3-layer cognitive architecture: reactive (classical), deliberate (lightweight LLM), reflective (daily/weekly reasoning)

Fee-Level Sweep

Vary the ToU schedule itself: the one experiment that would let us say how much to charge, not just whether

Scale Up

Scale beyond 2,500 agents ($\approx$ 400k vehicles today); dynamic OD matrix replacing the fixed assumption